

Predictive Modeling of Concrete Compressive Strength and Curing Age

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Note: This version of the report has been condensed for readability. A more comprehensive analysis, including full SAS code and output, is provided in the accompanying code report.

Abstract

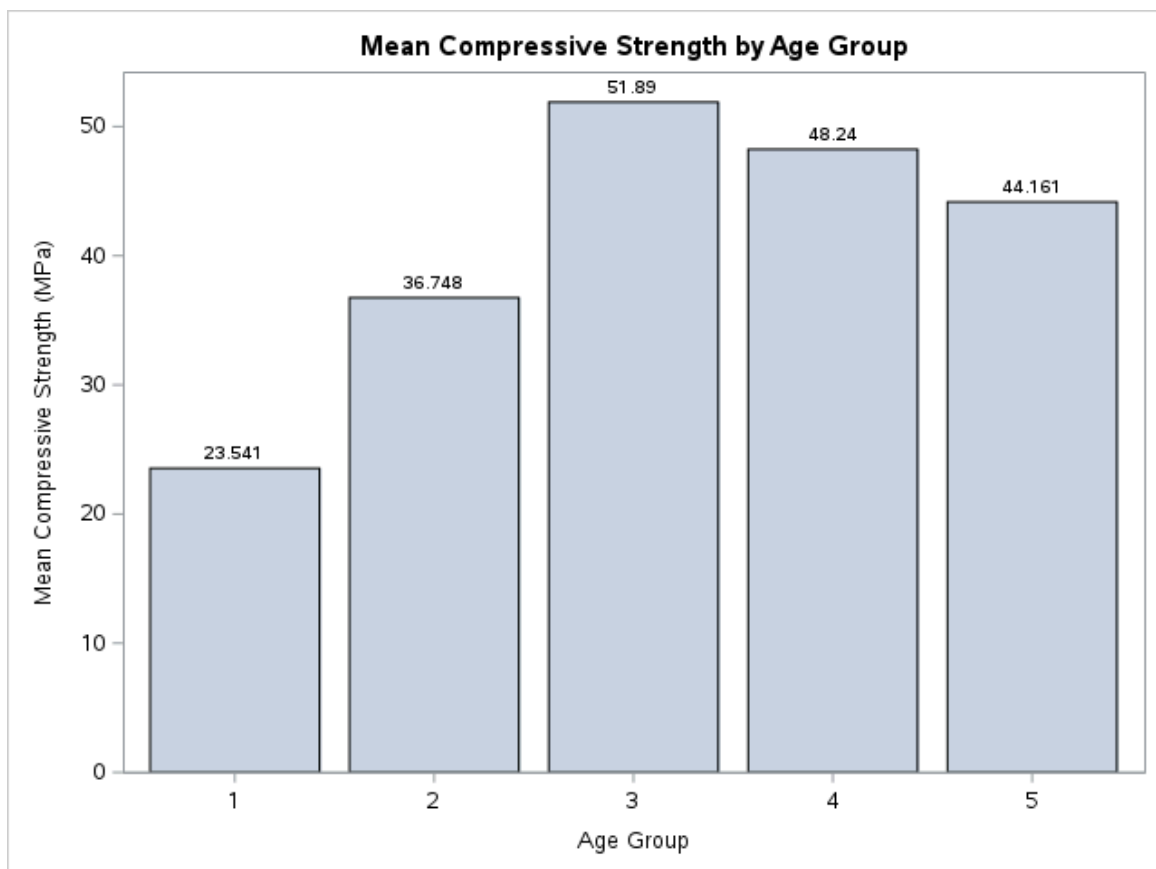
This report analyzes a dataset of concrete mix designs using SAS to help a construction manager make informed decisions about mixture composition, curing timelines, and quality control. Descriptive analysis confirmed that cement content and curing age are the two strongest indicators of compressive strength, which is consistent with Abrams' Law. In addition, there was no significant multicollinearity among the predictor group.

Three predictive models were developed to address specific client questions. First, a validated regression model estimates compressive strength for concrete cured at least 100 days, explaining about half of the variability in strength on the new, unseen data (out-of-sample $R^2 = 0.500$). Second, after verifying and correcting for a quasi-complete separation issue using Firth's penalized likelihood method, a logistic regression model reliably classifies whether concrete cured 90-100 days will reach a 50 MPa safety threshold, achieving 88.0% accuracy on a held-out test set. Third, an initial attempt to estimate concrete age from composition revealed a data leakage issue in the original modeling approach; once corrected, composition and strength explained a more modest 34% of age's variability, and quadratic discrimination analysis was used to classify concrete into five age groups with about 56% overall accuracy, performing most reliably for the oldest, most compositionally distinct concrete.

Throughout this analysis, several modeling issues, including data leakage, quasi-complete separation, and a violated proportional-odds assumption, were spotted through diagnostic testing and correction. All together, these findings provide the client with a practical foundation for determining mixture design, safety evaluation, and quality control decisions.

Section 1: Descriptive Statistics

We begin our analysis by examining this data's characteristics via descriptive statistics. We make sure to examine these various statistics to ensure that the data meets statistical assumptions that allows us to then interpret the data with accuracy. Based on the Normality assessment, all continuous predictors yielded $p < 0.01$ on formal normality tests, providing strong evidence against normality. It is worth noting that with a sample size of this scale ($n \sim 1030$), formal normality tests are very sensitive and will reject normality even for negligible deviations. The histograms and probability plots give a more interpretable picture of the actual distributional shape, as well as show that the variables are all right-skewed, consistent with our normality analysis. This analysis does make sense logically as concrete strength cannot be negative and a majority of the strengths are around a common range, typical strength level, with only a few high-strength outliers. For example, looking at fly ash and superplasticizer statistics, some observations have small amounts with others occasionally having higher amounts, resulting in producing the long right tails visualized in those variables graphics.



The Spearman correlation analysis revealed insight into how the component-to-water ratios, age, and compressive strength relate to one another. As the aforementioned Abrams' Law states, the cement-to water-ratio is strongly positively correlated with compressive strength ($\rho \sim 0.52$, $p < 0.0001$), while age shows an even stronger steady relationship ($\rho \sim 0.60$, $p < 0.0001$), confirming that both mixture composition and curing time are major contenders of concrete performance. The remaining component ratios display only weak to moderate correlations with strength, suggesting their effects are less influential predictors. Moderate positive correlations appear among several mix ratios. For instance, coarsewater and finewater ($\rho \sim 0.62$) and superplasticizer water with other components (ρ ranging from ~ 0.47 to ~ 0.58). These relationships are expected since certain components tend to be modified together in concrete production. Importantly, no pair of predictors exceeds an absolute correlation of 0.7, indicating no immediate signs of multicollinearity. Additionally, age is nearly uncorrelated with component ratios ($|p| < 0.09$), suggesting that curing time is largely independent of mix composition. Overall, the correlation pattern highlights cement content and age as the most influential variables for strength while showing that the set of predictors maintain a reasonable level of independence.

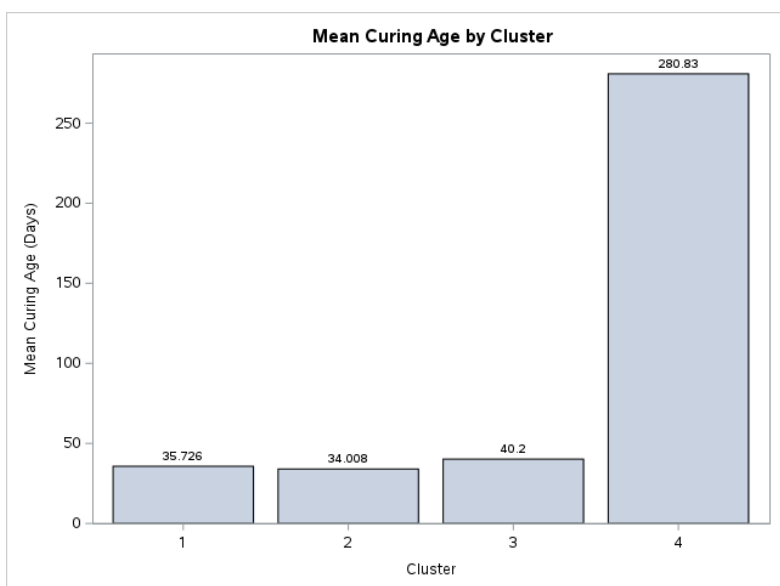
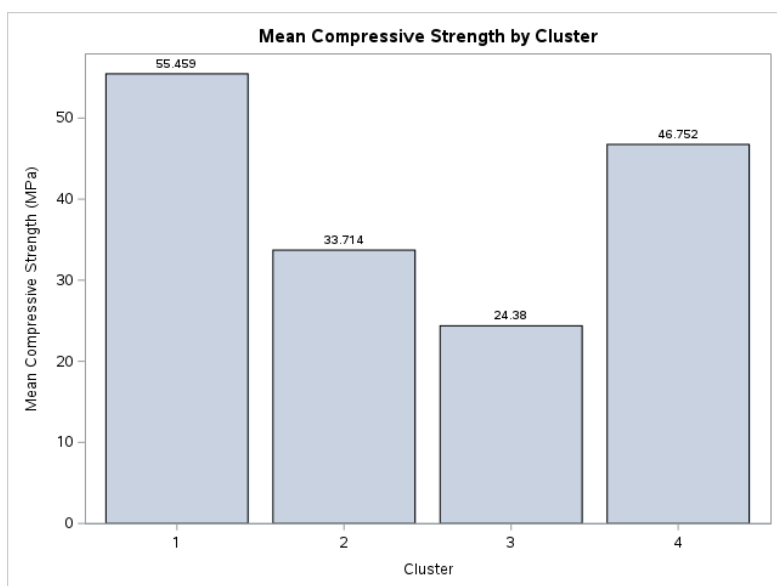
In order to examine how compressive strength develops over time, the concrete samples were split into 5 different age groups. As expected, compressive strength increases with curing age: the youngest concretes (<28 days, Group 1) have the lowest mean strength (23.54 MPa) and the largest variability (SD ~ 12.41 MPa), reflecting early-stages in strength development. Strength rises exceptionally in Group 2 (28-55 days, mean ~ 36.75 MPa). Group 3 (56-89 days), Groups 4 (90-179 days) and 5 (≥ 180 days) maintain relatively high strengths (51.89 MPa, 48.24 MPa and 44.16 MPa, respectively) with slightly lower variability, indicating that concrete strength stabilizes as it cures longer. The pattern confirms that both curing time and mixture composition strongly influence compressive strength, with strength increasing rapidly in early ages and leveling in older samples.

Note: the age groupings used for this descriptive analysis differ from the specific age thresholds used later in Section 3 (e.g., 100+ days, 90-100 days), which were defined for separate, more client-focused questions.

Section 2: Clustering

To explore natural groupings of concrete compositions and ages, hierarchical clustering was performed using the component-to-water ratios and age. Two procedural choices are worth noting: all variables were standardized prior to clustering, since the ratios and age are measured on very different metric scales and would otherwise let age dominate the grouping due to its larger numeric range; and compressive strength was deliberately withheld from the clustering variables themselves. The variable was added back only afterward to describe each group, so that the resulting clusters represent a genuine, independent pattern in mix composition and curing time rather than one that superficially 'explains' strength by construction.

Based on the pseudo F statistics, the analysis peaked at four clusters (134, compared to 89.2 at three clusters), and the pseudo t^2 statistic jumped sharply when merging from four to three clusters. These support a four-cluster solution as the best-choice grouping. Cluster 1 (n=73) represents a high-performing mix profile, combining high cement and slag content with minimal fly ash, achieving the greatest average compressive strength (55.46 MPa) at a curing age (≈ 36 days) comparable to the much larger dominant cluster. Cluster 2 (n=904) represents standard concrete mixes with lower cement content and moderate fly ash, averaging at a considerably lower strength (33.71 MPa) at a similar age. Cluster 3 (n=5) describes a more peculiar and small mixture with very high fly ash and aggregate content, as well as the lowest strength (24.38 MPa). Given its very small size, this probably reflects a handful of observations that are outliers rather than a specific, rare mixing strategy. Cluster 4 (n=48) is described by having high cement content, minimal additives, and a very long curing period (≈ 281 days), achieving the greatest overall strength (46.75 MPa).



These results suggest that composition, not just age, distinguishes great and poor-performing mixes even within similar curing periods. With regards to efficiency, cluster 1 in particular represents a proven strategy for achieving high early-to-moderate strength without the need of prolonged curing time.

Section 3: Compressive Strength for Long-Cured Concrete

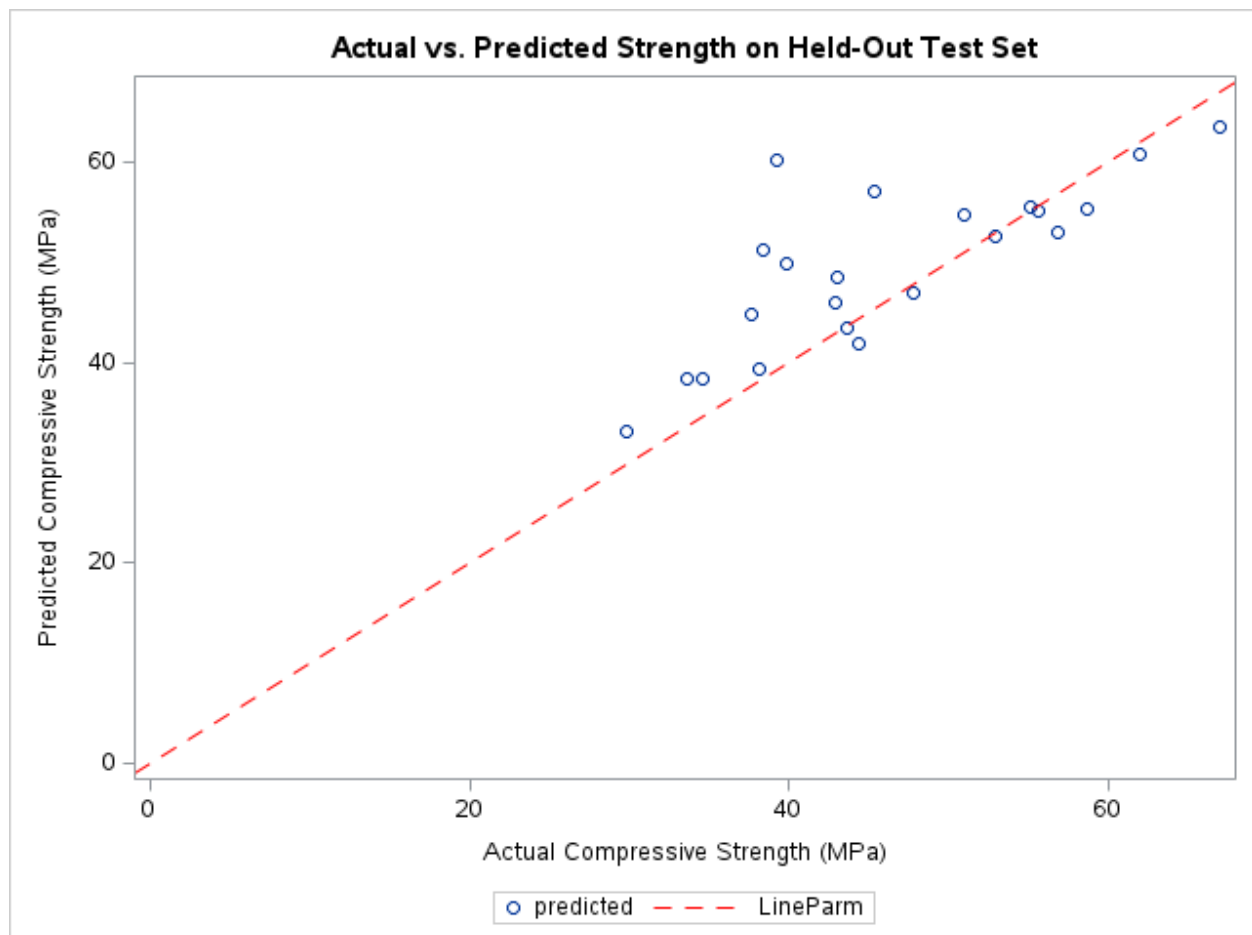
In this section, we are tasked with building various predictive models to provide useful insights for concrete performance and aging. Specifically, we aim to 1) predict the compressive strength of concrete that has cured for at least 100 days, 2) determine the likelihood that concrete cured between 90 and 100 days will achieve a strength of at least 50 MPa (for safety), and 3) estimate the approximate age of concrete samples based on their composition and observed strength data. These models will help the construction manager make informed decisions about concrete quality, curing timelines, and project efficiency. We will also assess model performance, identify the most influential factors, and highlight any limitations in predicting specific outcomes; we will begin with part (1).

Because compressive strength is a continuous variable, we apply a multiple linear regression model. While Section 1 identified non-normality in the raw predictor variables, regression validity depends on the normality of the model's residuals rather than the predictors themselves, so this was tested directly. One predictor, coarse-to-water ratio, was not statistically significant ($p = 0.399$) and showed an elevated variance inflation factor ($VIF = 10.56$). Given its lack of significance, it was removed going forward, consistent with its exclusion from the stepwise-selected model that followed. The residuals from the resulting model showed no notable stray from normality (Shapiro-Wilk $p = 0.102$; Kolmogorov-Smirnov, Cramér-von Mises, and Anderson-Darling tests all $p > 0.15$), all of which support the validity of this model's inference despite the skew present in the raw variables.

A stepwise regression model spotted four key predictors: superplasticizer-to-water, cement-to-water, slag-to-water, and fine-to-water ratios. The model explains about 56% of the variability in strength ($R^2 \approx 0.557$), and all variables are significant at the 0.05 threshold. Superplasticizer and cement ratios have the largest positive impact on strength, while slag contributes positively to a lesser extent; fine aggregate shows a small negative effect, suggesting

higher fine content may slightly reduce strength. Age did not enter the stepwise model, probably due to the sample already being restricted to concrete cured at least 100 days, leaving limited variation in age to detect a relationship.

In order to confirm that the model generalizes beyond the data it was fit on, the sample was split randomly into training and test sets (evaluated at both 70/30 and 80/20 splits) and the stepwise model was refit and re-evaluated on the withheld data. Cement, slag, and superplasticizer ratios were consistently chosen across both splits, and overall model fit remained close ($R^2 = 0.541$ and 0.534 for the two training splits, versus 0.557 for the full sample); however, the fourth variable selected differed between runs (fine-to-water ratio in one run, fly ash-to-water in the other). This is a known limitation of stepwise selection that is worth keeping in mind when interpreting which predictors are marked as most influential. Using the 80/20 split going forward, the refit model was evaluated on the withheld 20% test set, achieving an out-of-sample R^2 of 0.500 . This is a minor and expected decrease from the training and full sample results, and a good indicator that the model is not overfitting and performs arguably well on new data.

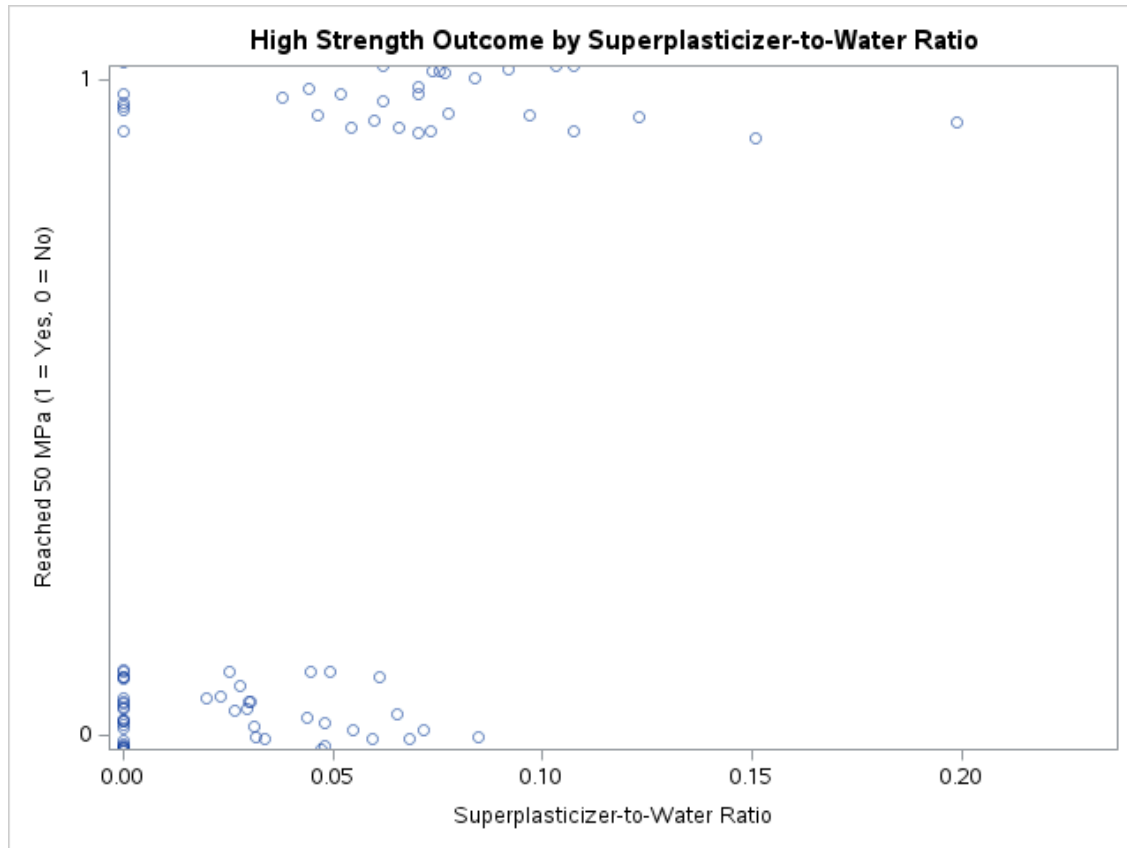


Based on this validation process, the recommended model for estimating compressive strength in concrete cured at least 100 days uses cement-to-water, slag-to-water, superplasticizer-to-water, and fly ash-to-water ratios as its predictors. This model explains roughly half of the variability in strength on new data, leaving a practical tool for estimating expected strength in older concrete based on its material composition.

Section 4: Safety Threshold Classification

For this task, the client requested an assessment of whether concrete cured within 90 and 100 days is likely to reach a compressive strength of 50MPa, a threshold which is important for reaching safety evaluation standards. Since the outcome is binary (i.e., reaching 50 MPa or not), logistic regression is appropriate, with the component-to-water ratios and age as predictors. Within the 90-100 day curing period, 128 concrete samples were accessible, of which 51 (39.8%) achieved the 50 MPa strength threshold and 77 (60.2%) did not. This sample size is acceptable and supported for the use of logistic regression going forward.

A full logistic model was first fit using all component-to-water ratios as predictors (age withheld from this initial model). Global tests confirm that the model significantly improves prediction over an intercept-only model, and the odds ratios show that increases in key components increase the likelihood of reaching 50 MPa strength. However, the odds ratios and confidence intervals for several predictors were extremely large, especially for variable superplasticizer-to-water ratio. A cross-tabulation of high-strength outcomes against superplasticizer-to-water ratio showed a quasi-complete separation pattern. At low values, almost all observations were below 50 MPa, and at high values, above it, with minimal overlap in the middle. This clarifies the unsteady odds ratios noticed earlier.



Note: Jitter was applied in the code in order to add minor vertical noise to the 0/1 points, avoiding them stacking into two flat horizontal lines and allowing the density of points at each value to be illustrated.

Stepwise selection, including age as an additional candidate predictor, spotted cementwater, slagwater, flyashwater, and superplasticizerwater as the most influential positive predictors. The age variable did not enter the model at any step, likely since there's a strict 11-day curing window. Influence diagnostics showed that no single observation had a concerning impact on the model, with the maximum Cook's distance under 0.7. The model's overall classification performance was strong (concordance = 95.1%), despite the Hosmer-Lemeshow test suggesting some misalignment between the predicted and observed probabilities ($\chi^2 = 16.63$, $p = 0.034$).

With regards to the separation identified above, Firth's penalized likelihood method was used, which stabilizes coefficient estimates even when a predictor almost perfectly sorts the results into pass or fail on its own. With this addition applied to the full six-predictor model, several odds ratios still remained extreme, indicating too little independent information across all

six predictors for the correction to fully stabilize each estimate. Due to Firth's correction not being able to be combined with stepwise selection within SAS, the three predictors already suggested as the most influential (cement-to-water, slag-to-water, and superplasticizer-to-water ratios) were used to fit a smaller, Firth-corrected model. This reduced model showed improvement in stability, with all three predictors remaining strongly, significantly positive.

In order to confirm this model generalizes to new data, the sample was split into training and testing sets, and the reduced Firth model was refitted on the training data and scored on the withheld 20% test set. The model successfully classified 22 of 25 test observations (88.0% accuracy), with 81.8% sensitivity and 92.9% specificity. This is a strong out-of-sample result, though the small test size means this should be treated as a promising indicator rather than a highly precise measurement.

Based on this validation process, the reduced three-predictor Firth model (using cementer-to-water, slag-to-water, and superplasticizer-to-water ratios) is recommended as the final tool for checking whether concremented cured during the 90-100 day period is likely to meet the 50 MPa safety threshold. For the client, the best takeaway is that increasing cement, slag, and superplasticizer content raises the odds of reaching this threshold; however, precise probability estimates should be examined with some caution given the separation issue identified prior.

Section 5: Estimating Concrete Age

For the final task, the goal is to estimate the approximate age of concrete samples based on their composition and observed strength data, classifying them into one of five age ranges specified by the client: under 28 days (Group 1), 28-55 days (Group 2), 56-89 days (Group 3), 90-179 days (Group 4), and 180 days or more (Group 5).

An initial linear regression of concrete component ratios, compressive strength, and age against the derived age-group variable seems to explain 72% of the variability ($R^2 = 0.721$). However, upon closer inspection, it was revealed that this result was driven by data leakage. Since age was included as a predictor of a variable derived from itself (age-group), this circular logic inflated the R^2 value. Once corrected using continuous age from composition and compressive strength, excluding age as a predictor, the model explains 34% of the variability in age ($R^2 = 0.343$). Compressive strength is a strong significant positive predictor of age ($t = 20.36$, $p < 0.0001$), which is congruent with the expectation that concrete becomes stronger over

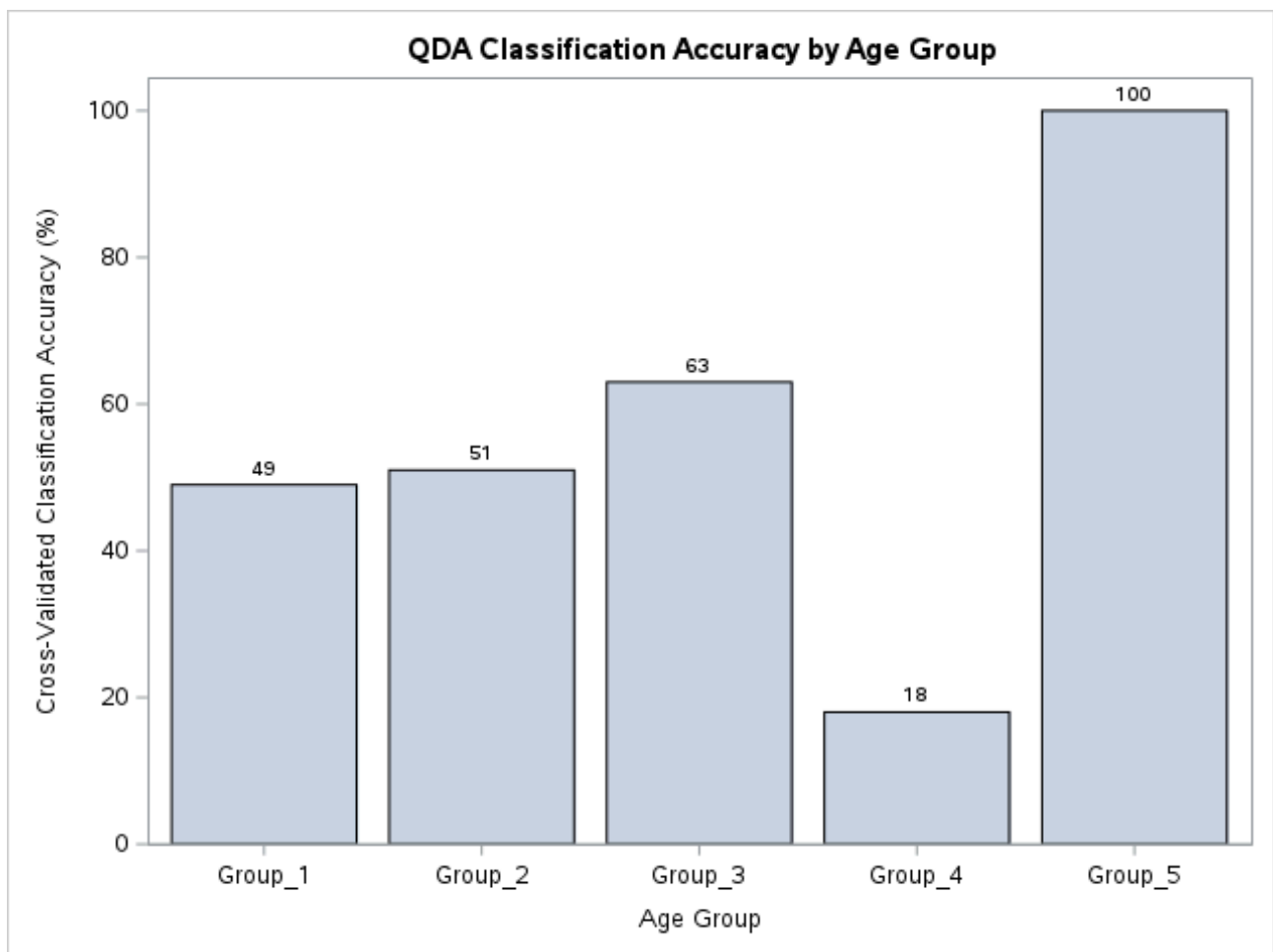
time. Cement-to-water, slag-to-water, and fly ash-to-water ratios were also significant, but showed a negative association with age. This is most likely the result of differences in mix design and intentions, for example, some mixes may be created for faster projects, while others require long-term curing. Variable superplasticizer-to-water and coarse-to-water ratios were not significant. All variance inflation factors remained well below 10 (maximum 3.12), indicating no multicollinearity issues.

To further confirm this leakage concern, a logistic model with age as a predictor of the age-group variable was tested and immediately returned with complete separation of data points. This is a clear illustration that age alone inherently determines the age group by construction. After removing age and including compressive strength along with the composition ratios, the logistic model converged normally and yielded a discrimination value of 0.890. Despite this, two concerns remain: 1. The Hosmer-Lemeshow test suggests lack of fit ($\chi^2 = 113.27$, $p < 0.0001$), and 2. The proportional odds assumption was significantly violated ($\chi^2 = 312.10$, $p < 0.0001$). Due to this, discriminant analysis was explored as an alternative means to bypass this proportional odds assumption.

Before choosing between linear (LDA) and quadratic (QDA) discriminant analysis, a test was run to verify whether the covariance structure of the predictors was consistent across all five age groups. They were not consistent ($\chi^2 = 2725.66$, $df = 112$, $p < 0.0001$), backing the use of QDA, since it allows each group to have its own covariance structure. Out of the groups, age group 5 stood out as its covariance matrix had a lower rank than the other groups. This was linked back to its two predictors, fly ash-to-water and superplasticizer-to-water ratios, which both showed zero variance within this group (all 59 observations used were exactly 0 for each). Seemingly, concrete mixes within this age group (cured 180+ days) consistently used no fly ash or superplasticizer whatsoever which is very distinct from the other age groups. This is consistent with Cluster 4 noted in Section 2, describing a similar long-cured, fly ash-free, and high-cement profile.

The quadratic discriminant analysis demonstrated a statistically significant ability to distinguish the five age groups (Wilks' Lambda, Pillai's Trace, Hotelling-Lawley Trace, and Roy's Greatest Root all $p < 0.0001$). The cross-validated classification results show that QDA correctly classifies a good portion of observations in some age groups. Performance-wise, age groups 1, 2, and 3 show moderate accuracy (49%, 51%, and 63%, respectively). Age group 4

was considerably more difficult to distinguish (18% accuracy). Age group 5 was classified correctly 100% of the time, a result of its distinct, zero fly-ash composition rather than its small sample size. Stepwise discriminant analysis verified compressive strength as the single most influential predictor for distinguishing age groups (partial $R^2 = 0.338$), followed by cement-to-water and slag-to-water ratios. Overall, the cross-validated misclassification rate is approximately 44%, suggesting that while the model does capture general trends, there is notable overlap amongst groups 1, 2, and 4, a limitation of the current set of variables for precise age classification.



Conclusion

This analysis examined how concrete analysis confirmed that composition and curing age relate to compressive strength, and developed a set of predictive models to support the client's decision-making around mixture design, safety thresholds, and quality control.

Descriptive and correlation analysis proved that cement content and curing age are the two greatest drivers of compressive strength, which is consistent with Abrams' Law. The predictor group overall did not show any major signs of multicollinearity concerns. Clustering pinpointed four distinct mix profiles, most notably a high-cement, high-slag, low-fly-ash, mixture (Cluster 1) which achieved strength akin to the longest-cured mixtures at a fraction of the curing age.

For checking whether concrete cured within a 90-100 days window reaches the 50 MPa safety threshold, an initial logistic model revealed quasi-complete separation driven by superplasticizer-to-water ratio. After applying Firth's penalized likelihood correction and validating the held-out test set, the final three-predictor model achieved a strong out-of-sample performance (88.0% accuracy), supporting cement, slag, and superplasticizer content as key components of reaching the strength threshold. However, the probability estimates should be interpreted with slight caution given the separation detected.

For estimating concrete age, an initial regression revealed data leakage from having age-group as the response alongside age as a predictor variable. Once corrected, composition and strength explained a more modest but valid 34% of age's variability. A parallel attempt at logistic classification of age groups confirmed the same leakage issue through complete separation. After removing age, it showed reasonable discrimination but still violated a key modeling assumption. Quadratic discriminant analysis was then chosen since it does not rely on that assumption, and it classified age groups with an overall accuracy of roughly 56% (44 misclassification), performing the best for age groups 3 and 4 and most reliably for age group 5. Age group 5's distinct composition consisting of fly-ash-free and long-cured quality falls in line with Cluster 4 from the clustering analysis section, also making it more easily distinguishable from other groups.

To conclude everything, these analyses confirm that both mixture composition and curing time purposefully influence concrete performance, despite their relative importance varying by task. Composition dominates in differentiating high-performing mixes and safety thresholds, while age remains only somewhat predictable based on composition and strength alone. Many models required correcting for methodological issues identified during analysis, including data

leakage, quasi-complete separation, and violated model assumptions, all of which emphasize the importance of diagnostic validation before choosing any given model's results. The corrected, validated models presented here provide the client with a practical, evidential foundation for mixture design, safety evaluation and quality control decisions.